

Business Problem Statement

I chose to explore this problem because customer retention and purchase behavior are some of the most common challenges retail businesses face, and I wanted a dataset rich enough to practice the full analyst workflow -- from raw data to a decision-ready dashboard.

A mid-sized retail business is trying to understand why some customers keep coming back while others buy once and disappear. Management has noticed shifts in buying patterns across age groups, product categories, and channels, but doesn't have a clear, data-backed view of what's actually driving those shifts -- discounts, reviews, seasonality, or payment preferences. I set out to answer this for them using the customer shopping dataset:

"What patterns in customer behavior can this business act on to increase repeat purchases, improve loyalty, and spend its marketing budget more effectively?"

My Approach

I treated this as a full end-to-end analyst project rather than a single script, structured around four stages:

- **Data Preparation (Python):** Clean the raw dataset, handle missing values, and engineer new features (age bands, an adapted recency signal) so it's ready for analysis.
- **Analysis (SQL):** Load the cleaned data into PostgreSQL and write queries to segment customers, uncover purchase drivers, and answer key business questions.
- **Visualization (Power BI):** Turn the SQL output into an interactive dashboard so the findings are usable by someone without a SQL background.
- **Reporting:** Summarize findings into a short report and a set of business recommendations a retail team could realistically act on.

What I Wanted to Get Out of This

- Practice writing SQL beyond basic aggregates -- window functions and CTEs for scoring and ranking.
- Get comfortable moving data from a pandas pipeline into a real relational database.
- Build a dashboard meant for a non-technical stakeholder to actually use, not just display charts.
- Turn analysis into recommendations a business could act on, not just descriptive statistics.

Deliverables

- Cleaned dataset and Python notebook covering EDA, cleaning, and feature engineering.
- SQL script with business-question queries, including the RFM segmentation logic.
- Interactive Power BI dashboard.

- Short written report and summary of recommendations.
- GitHub repository with all of the above, organized and documented.